END-TERM EXAMINATION, SPRING, 2022

CO216 FORMAL LANGUAGES AND AUTOMATA

Max. Marks: 60 **Max Time: 3 hrs**

Answer all questions

1. State **TRUE** or **FALSE** of the following statements: 1X 8 = 8

   a) Recursive and recursively enumerable languages are the same class of languages.
   b) Turing Machine may halt even when the given input string does not belong to the given language.
   c) A Turing Machine with five input tapes is more powerful than a single-tape Turing Machine.
   d) For every problem in the world, there exists a Turing Machine to solve it.
   e) Halting Turing Machines accept recursive languages.
   f) The language $L = \{a^n b^n c^n, \Sigma = \{a,b,c\}, n>0\}$ is a Context-Sensitive Language.
   g) The language $L = \{ww^R | w \in \{a,b\}^+\}$ can be accepted by a Deterministic Turing Machine.
   h) Universal Turing Machine is a Multi-tape Turing Machine.

2. Fill up the blanks: 1 X 4 = 4

   a) A function *f* is ________ if there exists a __________.
   b) Context-Sensitive Language class is a _______ of Recursive Language class.
   c) For every ________ _________ language there exists an unrestricted grammar that generates the language.
   d) In case of __________ _______ grammar leftmost and rightmost derivations are the same.

3. Answer the following in short 4X3 = 12

   a) What are normal forms of a Context-Free Grammar? What are the characteristics and advantages of normal forms?
   b) Transform the grammar with productions $S \rightarrow aSaA \mid A;\ A \rightarrow abA \mid b$ into Chomsky Normal Form
   c) Convert the grammar $S \rightarrow ABb \mid a \mid b;\ A \rightarrow aaA \mid B;\ B \rightarrow bAb$ into Greibach Normal Form.

4. (a) Define a Mealy Machine. How does it differ from a Moore Machine? How to convert a Mealy machine to an equivalent Moore Machine? 6

   (b) Design a Mealy Machine that computes 2's complement of a binary number given as input. Also, convert it to its equivalent Moore machine. 6

5. (a) Define a Linear-Bound Automata. Explain how a Linear-Bound Automata Works. 6

   (b) Design a Linear-Bound Automata that accepts the language $L=\{a^n b^n c^n, \Sigma=\{a,b,c\}, n>0\}$ 6

6. (a) Define a Turing Machine. Explain how a Turing Machine works. 6

   (b) Design a Turing Machine that accepts the language $L = \{ww^R \mid w \in \{a, b\}+\}$. 6